

SUBIN LIN

+65 83014240 | *Singapore*

necolsy@gmail.com | [GoogleScholar](#) | [ORCID](#)

EDUCATION

PhD in Built Environment	2023.01 - expected 2026.12
National University of Singapore (NUS)	
MSc by Research in Civil and Environmental Engineering	2020.08 - 2022.06
Korea Advanced Institute of Science and Technology (KAIST), GPA: 96.67/100	
BSc in Civil and Environmental Engineering	2016.08 - 2020.08
Technion - Israel Institute of Technology, GPA: 89.3/100	

RESEARCH FOCUS

PhD candidate combining wind-tunnel-validated CFD, RANS/LES model assessment, and applied AI/ML using PyTorch, graph neural networks, multivariate time-series forecasting, and physics-informed LLMs. Interested in developing physics-informed AI and surrogate modelling methods to accelerate CFD-based flow prediction, validation, and design optimisation while reducing computational cost.

SELECTED PUBLICATIONS AND PREPRINTS

Journal Articles

- **S. Lin**, J. Leong, H. J. Poh. Design trade-offs in building porosity: A parametric analysis of vertical placement and geometry for urban ventilation. *Journal of Wind Engineering and Industrial Aerodynamics*, 271, 106369, 2026. [doi](#)
- **S. Lin**, V. Ramani, M. Martin, P. Arjunan, A. Chong, F. Biljecki, M. Ignatius, K. Poolla, C. Miller. District-scale surface temperatures generated from high-resolution longitudinal thermal infrared images. *Scientific Data*, 10, 859, 2023. [doi](#)
- **S. Lin**, J. Kim, C. Hua, S. Kang, M. H. Park. Comparing artificial and deep neural network models for prediction of coagulant amount and settled water turbidity: Lessons learned from big data in water treatment operations. *Journal of Water Process Engineering*, 54, 103949, 2023. [doi](#)
- **S. Lin**, J. Kim, C. Hua, M. H. Park, S. Kang. Coagulant dosage determination using deep learning-based graph attention multivariate time series forecasting model. *Water Research*, 232, 119665, 2023. [doi](#)

Conference and Workshop Papers

- **S. Lin**, J. Leong, B. C. Khoo, H. J. Poh. Aerodynamic response and ventilation enhancement in urban street canyons through building porosity: A comparative numerical and experimental study. *E3S Web of Conferences*, 716, 10003, 2026 (IAQVEC 2026). [doi](#)
- **S. Lin**, C. Hua. Physics-Informed Large Language Models for HVAC Anomaly Detection with Autonomous Rule Generation. *NeurIPS 2025 Workshop on UrbanAI*, Oral, 2025. [arXiv](#)
- **S. Lin**, J. Leong, H. J. Poh. The role of building porosity in enhancing passive ventilation in high-density urban street canyons. *6th International Conference on Building Energy and Environment (COBEE 2025)*, July 2025.
- **S. Lin**, C. Hua. BuildEvo: Designing Building Energy Consumption Forecasting Heuristics via LLM-driven Evolution. *ICML 2025 Workshop*, Poster, 2025. [arXiv](#)
- K. Eum*, V. Erb*, **S. Lin***, S. Wang*, Y. Y. Doh. How the death-themed game *Spiritfarer* can help players cope with the loss of a loved one. *CHI '21 Extended Abstracts*, 2021. *Equal contribution. [doi](#)

RESEARCH EXPERIENCE

Wind-tunnel validated CFD for porous urban street canyons 2025.08 - present
Independently designed and conducted pressure-tap wind-tunnel experiments for porous street-canyon models, including model preparation, tubing/DAQ setup, pressure-transducer calibration, and uncertainty analysis. Validated CFD pressure predictions across five inlet velocities, with detailed cases achieving Pearson correlations above 0.97, and generated 20 CFD cases for porosity-geometry, vertical-placement, pedestrian-level mean-flow, and ACH analysis.

RANS and LES fidelity assessment for urban ventilation design 2025.09 - present

Compared RANS and selected LES simulations under pressure-tap-only validation constraints. Assessed mesh sensitivity, inlet-turbulence sensitivity, LES averaging effects, and computational cost, establishing a validation-aware CFD basis for future surrogate modelling and model-fidelity selection.

AI/ML modelling for environmental and building systems 2021.08 - present

Built a solid applied AI/ML foundation through PyTorch-based graph attention and recurrent neural network models for multivariate operational forecasting in drinking-water treatment, and AI frameworks for built-environment prediction and diagnostics, including BuildEvo for interpretable building-energy forecasting heuristics and PILLM for physics-informed HVAC anomaly detection.

CFD-based aircraft air-terminal-device design 2025.03 - 2026.03

Conducted Fluent-based CFD simulations for personalized air terminal device (ATD) design in aircraft cabins. Compared outlet geometries using performance criteria, supported 3D printing of selected designs, and contributed to experimental validation planning.

Thermal-infrared urban dataset and analytics 2022.06 - 2022.12

Built Python workflows for district-scale longitudinal thermal-infrared image processing and urban heat analysis, contributing to a published *Scientific Data* dataset paper.

AI-assisted CFD and data-analysis workflows 2025.03 - present

Built Python workflows for CFD case organization, post-processing, validation metrics, and research analysis, with an emphasis on reducing manual simulation-analysis effort and preparing CFD outputs for data-driven modelling.

FUNDED RESEARCH PROJECTS

NUS-SIA Corporate Laboratory Project 2 2025.03 - 2026.03

Collaborated on aircraft-cabin air-terminal-device CFD research supported by the Agency for Science, Technology and Research (A*STAR) through grant A-0003501-08 and Project 2 of the NUS-SIA Corporate Laboratory.

NRF Korea drinking-water treatment project 2022.03 - 2022.08

Collaborated on *Big data and deep learning modeling of optimal coagulant dosage for reducing membrane fouling and DBP formation in drinking water treatment plants*, funded by the National Research Foundation of Korea (2022R1A2C100924211, PI: M. H. Park).

SELECTED CROSS-DISCIPLINARY PROJECTS

CHI EA '21 paper on serious games and bereavement 2020.10 - 2021.02

Rapidly learned human-computer interaction and qualitative research methods, conducted interviews and thematic analysis on how the game *Spiritfarer* supports players in coping with loss, and published in *CHI EA '21* as a co-first author.

AI skin-analysis application 2021.11

Implemented an AI coding prototype for skin-lesion image diagnosis using HAM10000 image inputs, OpenCV preprocessing, U-Net segmentation, DenseNet-based classification, and app/web interfaces. Awarded third prize in the AI & Big Data Hackathon by Daewoong Foundation.

TECHNICAL SKILLS

CFD and experimental validation: ANSYS Fluent, OpenFOAM, ParaView, RANS/LES, pressure-tap wind-tunnel experiments, uncertainty analysis, CFD validation, ACH and pedestrian-level ventilation metrics.

Scientific computing and machine learning: Python, PyTorch, graph neural networks, recurrent networks, multivariate time-series forecasting, surrogate modelling, physics-informed AI, LLM-based diagnostics, and AI-assisted CFD workflow automation.

WORK EXPERIENCE

Senior Research Engineer 2022.09 - 2022.12

Berkeley Education Alliance for Research in Singapore Limited

Supported an infrared thermography field campaign, data-quality checks, Python analysis workflows, and manuscript preparation for built-environment sensing research.

TEACHING EXPERIENCE

Teaching Assistant, BCA-GMAAP Certification Course	2026.06
CFD Modelling for Natural Ventilated Buildings	
Supported professional training on CFD fundamentals, natural-ventilation modelling, wind-driven rain, verification/validation, and hands-on CFD tutorials.	
Teaching Assistant, BPS5112 Green Building Integration and Evaluation Studio	2024.08
National University of Singapore	
Supported teaching on natural ventilation modelling using CFD for built-environment graduate students.	

HONORS AND AWARDS

IAQVEC 2026 Fellowship (Participation Support), IAQVEC 2026	2026.05
A*STAR Merit Award (AMA, SINGA), A*STAR	2022.12
Third Prize, AI & Big Data Hackathon, Daewoong Foundation	2021.11
Full 2-year Scholarship, KAIST	2020.08 - 2022.05
Dean's List with Scholarships, Technion	2018/2019 spring and winter semesters
Full 4-year Scholarship, Technion	2016.08 - 2020.08